RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-01

Single Correct Question

1. The number of different orders of a matrix having 12 elements is

(A) 3 (B) 1 (C) 6 (D) None of these

MMT234

2. Construct a 3×2 matrix whose element are given by $a_{ij} = \frac{1}{3}|2i + j|$

(A) $\begin{bmatrix} 1 & \frac{5}{3} \\ \frac{4}{3} & 2 \\ \frac{7}{3} & \frac{8}{3} \end{bmatrix}$ (B) $\begin{bmatrix} 1 & \frac{4}{3} \\ \frac{5}{3} & 2 \\ \frac{7}{3} & \frac{8}{3} \end{bmatrix}$ (C) $\begin{bmatrix} 1 & \frac{4}{3} \\ \frac{5}{3} & 2 \\ \frac{8}{3} & \frac{7}{3} \end{bmatrix}$ (D) none of these

MMT235

3. Which of the following statement are true:

1. Every identity matrix, I of order n is an upper triangular matrix.
2. Every identity matrix, I of order n is a lower triangular matrix.

(A) Only 1 (B) Only 2 (C) Both 1 and 2 (D) None of these

Rankers Academy JEE

MMT236

4. Consider the following statements:

1. Every zero matrix is a square matrix.
2. A matrix has a numerical value.
3. A unit matrix is a diagonal matrix.

Which of the above statements is/are correct?

(A) 2 Only (B) 3 Only (C) 2 and 3 (D) 1 and 3

MMT237

5. $\begin{bmatrix} x^2 + x & x \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ -x + 1 & x \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 5 & 1 \end{bmatrix}$ then x is equal to

(A) – 1 (B) 2 (C) 1 (D) No value of x

MMT238

6. If $2A + B = \begin{bmatrix} 6 & 4 \\ 6 & -11 \end{bmatrix}$ and $A - B = \begin{bmatrix} 0 & 2 \\ 6 & 2 \end{bmatrix}$, then A =

(A) $\begin{bmatrix} -2 & 2 \\ 4 & -3 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & 2 \\ 4 & -3 \end{bmatrix}$ (D) none of these

MMT239

7. If A and B are square matrices of order 3 such that $|A| = -1, |B| = 3$, then $|3AB|$ is equal to -

(A) – 9 (B) – 81 (C) – 27 (D) 81

MMT240

For More Join: @IITJEE_Advanced

8. If $\begin{bmatrix} 2x+y & 2 & x-2y \\ a-b & 2a+b & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & -1 & -3 \end{bmatrix}$, then find the value of $x + y + a + b$

(A) 1 (B) –1 (C) –3 (D) 5

MMT241

9. Find the value of x : $2\begin{bmatrix} 3 & 1 & -2 \\ -1 & -3 & 4 \end{bmatrix} + x\begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -2 \\ -8 & -14 & -2 \end{bmatrix}$

(A) –2 (B) 2 (C) 3 (D) –3

MMT242

10. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$, then

(A) $AB = \begin{bmatrix} -5 & 8 & 0 \\ 0 & 4 & -2 \\ 3 & -9 & 6 \end{bmatrix}$ (B) $AB = \begin{bmatrix} -2 & -1 & 4 \end{bmatrix}$

(C) $AB = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ (D) AB does not exist

MMT243

11. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $j = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ then B equal to

(A) $I\cos\theta + j\sin\theta$ (B) $I\cos\theta - j\sin\theta$ (C) $I\sin\theta + j\cos\theta$ (D) $-I\cos\theta + j\cos\theta$

MMT244

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	C	B	A	C	B	B	A	D
Question	11									
Answer	A									

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-02

Single Correct Question

1. If $A=\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, where $i=\sqrt{-1}$ and $x \in N$, then A^{4x} equals -

(A) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$

MMT245

2. For $\theta=\frac{3\pi}{5}$, let $B=\left[b_{ij}\right]$ be a square matrix of 2nd order such that

$$b_{ij}=\begin{cases} \cos\theta & , \quad i=j \\ \cos\left(\frac{j\pi}{2}+\theta\right) & , \quad i>j \\ \sin\left(\frac{j\pi}{2}-\theta\right) & , \quad i<j \end{cases},$$

then trace of matrix B^5 is equal to -

(A) –2 (B) –1 (C) 1 (D) 2

Rankers Academy JEE

MMT246

3. If A and B are 3 × 3 matrices such that $AB = A$ and $BA = B$, then-

(A) $A^2 = A, B^2 \neq B$ (B) $A^2 \neq A, B^2 = B$

(C) $A^2 = A, B^2 = B$ (D) $A^2 \neq A, B^2 \neq B$

MMT247

4. If A, B are two square matrices such that $AB = A$, $BA = B$ then $(A + B)^7$ equals

(A) $A + B$ (B) $2^7(A + B)$ (C) $2^6(A + B)$ (D) $2^8(A + B)$

MMT248

Integer Type Question

5. Let A be a square matrix satisfying $A^2 = 2I$ and $\sum_{k=1}^{2n} A^k = 63A(A+I)$, then the value of n is

(where I denotes identity matrix)

MMT249

6. If $A=\begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $I=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find k so that $A^2 = kA - 2I$.

MMT250

For More Join: @IITJEE_Advanced

7. Let $A=\begin{bmatrix}\alpha^2(a-2\lambda) & 0 & 3\\ -1 & \alpha(b-3\lambda) & 1\\ 0 & 5 & (c-4\lambda)\end{bmatrix}$; $B=\begin{bmatrix}\beta^2(a-2\lambda) & -5 & -3\\ 7 & \beta(b-3\lambda) & 7\\ 8 & 9 & (c-4\lambda)\end{bmatrix}$ and

$C=\begin{bmatrix}\gamma^2(a-2\lambda) & -1 & 3\\ 0 & \gamma(b-3\lambda) & 7\\ 8 & -3 & (c-4\lambda)\end{bmatrix}$, where α, β, γ are real & distinct such that

$tr(A) = tr(B) = tr(C) = -1$, where $\lambda \in N$, then minimum possible value of $a + b + c$ is

MMT251

8. Let $A=\begin{bmatrix}-3 & 0 & 2\\ 1 & x & 5\\ -2 & 0 & x^2\end{bmatrix}, B=\begin{bmatrix}2\\ b\\ -1\end{bmatrix}$ and $C = [3\ \ 5\ \ 1]$, then the number of integral value(s) of 'b' for which

$Tr(ABC) \leq -18\ \forall\ x \in R$ is (are)

MMT252

Subjective Type Question

9. If $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$, then using principle of mathematical induction or otherwise prove that $A^n=\begin{bmatrix}\cos n\alpha & \sin n\alpha\\ -\sin n\alpha & \cos n\alpha\end{bmatrix}$ where n is a positive integer.

Rankers Academy JEE

MMT253

10. If $A=\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$, show that $A^2 = 3A$ and $A^4 = 27A$. Deduce that $A^3 - A = 2A(A + I)$

MMT254

11. **(a)** $A=\begin{bmatrix}2 & 3\\ 5 & -1\\ 7 & 8\end{bmatrix}$ then write A^T, **(b)** $A=\begin{bmatrix}5 & 1 & 7\\ 6 & 8 & 9\end{bmatrix}$ and $B=\begin{bmatrix}2 & 3\\ 1 & 2\\ 5 & 1\end{bmatrix}$ then find $A + B^T$.

MMT255

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	C	A	C	C	6	1	8	5
Question	11							
Answer	(a) $\begin{bmatrix}2 & 5 & 7\\ 3 & -1 & 8\end{bmatrix}$ (b) $\begin{bmatrix}7 & 2 & 12\\ 9 & 10 & 10\end{bmatrix}$							

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-03

Single Correct Question

1. The matrix $A=\begin{bmatrix}1 & 2 & 3\\ 1 & 2 & 3\\ -1 & -2 & -3\end{bmatrix}$ is

(A) idempotent matrix (B) involutary matrix

(C) nilpotent matrix (D) periodic matrix

MMT256

2. If $A, B, A+I, A+B$ are idempotent matrices (where A and B are square matrices of same order & I is identity matrix of same order), then value of $|AB-BA|$ (where $|C|$ denotes determinant of matrix C) is -

(A) 0 (B) 1 (C) 2 (D) cannot be determined

MMT257

3. If A is nilpotent matrix of index 2, then $A(I_2+A)^{51}$ is equal to (where I_2 is the identity matrix of order 2) -

Rankers Academy JEE

(A) A^{51} (B) I_2 (C) Null matrix (D) A

MMT258

4. If $A=\begin{bmatrix}a & b\\ c & 1-a\end{bmatrix}$ is idempotent matrix and $bc = 1/4$ then the value of a is

(A) 1 (B) –1 (C) 1/2 (D) –1/2

MMT259

5. If $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ \frac{-1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}$, then $(BB^TA)^5$ is equal to

(Where X^T denotes transpose of matrix X)

(A) $\begin{bmatrix}2+\sqrt{3} & 1\\ -1 & 2-\sqrt{3}\end{bmatrix}$ (B) $\frac{1}{2}\begin{bmatrix}1 & 5\\ 0 & 1\end{bmatrix}$ (C) $\begin{bmatrix}1 & 5\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}5 & 1\\ 0 & 1\end{bmatrix}$

MMT260

6. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $|A + B|$ is -

(A) 0 (B) –1 (C) 1 (D) 2

MMT261

For More Join: @IITJEE_Advanced

Multiple Correct Question

7. If A and B are square matrices such that $AB = B$ and $BA = A$, then which of the following is/are always true ?

(A) A is an idempotent matrix
(B) B is an involutary matrix
(C) $A^6 + B^6 = A^8 + B^8$
(D) $A^2 + B^2 = A + I$

MMT262

Integer Type Question

8. Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Let $n(A)$ denotes the number of elements in A and $n(XY) = \mathbf{0}$, when the two matrices X and Y are not conformable for multiplication. If $C = (AB)(B'A)$; $D = (B'A)(AB)$ then, find the value of $\left(\frac{n(C)\left(\left(|D|^2 + n(D)\right)\right)}{n(A) - n(B)} \right)$

MMT263

Subjective Type Question

9. If the matrix A is involutary, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A).\frac{1}{2}(I - A) = 0$.

MMT264

Rankers Academy JEE

10. Show that the matrix $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ is an orthogonal matrix.

MMT265

11. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying $AA^T = 9I$ where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to

MMT266

ANSWER KEY

Question	1	2	3	4	5	6	7	8	11
Answer	C	A	D	C	C	C	A,C	650	(–2, –1)

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-04

Single Correct Question

1. If A is a matrix of order 3 × 3 such that $A^2 + A = A^T$, then which of the following is wrong -

(A) $|A| = 0$ (B) $|A| \neq 0$

(C) A^2 is a skew symmetric matrix (D) A is singular matrix

MMT267

2. Let A and B be 3 × 3 matrices of real number where A is symmetric, B is skew symmetric and $(A + B)(A - B) = (A - B)(A + B)$ if $(AB)^T = (-1)^k AB$, then possible value of k is -

(A) 3 (B) 4 (C) 6 (D) 2

MMT268

3. If $A=\begin{bmatrix}1 & 0 & 1\\ 0 & 2 & 2\\ 1 & 1 & 3\end{bmatrix}$, then $tr(A^3 adj(A^2))$, is

(where $tr(A)$ and $adj(A)$ denote trace and adjoint of matrix A) -

(A) 6 (B) 24 (C) 36 (D) 12

MMT269

4. If A is a square martix of order n with $|A| = 9$ and $|adj\ (adj\ (adjA))| = 3^{16}$, then number of elements in A, is -

(A) 9 (B) 4 (C) 5 (D) 2

MMT270

Bankers Academy JEE

Multiple Correct Question

5. Let $A=\begin{bmatrix}0 & \tan\theta & 2\\ -\tan\theta & 0 & \sec\theta\\ -2 & -\sec\theta & 0\end{bmatrix}$, where $\theta \neq (2n+1)\frac{\pi}{2}, (n \in I)$ & $B=\sum_{r=1}^{5} A^{(2r-1)}$ then which of the following statement is correct

(A) A is skew symmetric matrix (B) B is skew symmetric matrix

(C) $(A + B)$ is skew symmetric matrix (D) $(A + B)$ is symmetric matrix

MMT271

6. Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j,\ x \in R \\ 1 & \text{if } |i-j| = 1 \\ 0 & \text{otherwise} \end{cases}$, then which of the following hold(s) good ?

(A) for $x = 2$, A is a diagonal matrix.

(B) A is a symmetric matrix

(C) for $x = 2$, det A has the value equal to 6

(D) Let $f(x) = \det A$, then the function $f(x)$ has both local maxima and local minima.

MMT272

For More Join: @IITJEE_Advanced

Subjective Type Question

7. If A and B are non-singular matrices of order three such that $adj(AB)=\begin{bmatrix}1 & 1 & 1\\ 1 & p & 1\\ 1 & 1 & p\end{bmatrix}$ and $|B^2\ adj(A)| = p^2 - 3$, then p is

MMT273

8. $A_{3\times3}$ is a matrix such that $|A| = a$, $B = (adj\ A)$ such that $|B| = b$. Find the value of $(ab^2 + a^2b + 1)S$

where $\frac{1}{2}S = \frac{a}{b}+\frac{a^2}{b^3}+\frac{a^3}{b^5}+.....$ up to ∞, and $a = 3$.

MMT274

9. If A be a square matrix of the order 5 and $B = Adj(A)$ then find $Adj\ (5A)$.

MMT275

10. If A be a square matrix of order 4 and $|A| = 3$ then find $adj(adjA)$.

MMT276

11. Express matrix $A=\begin{bmatrix}2 & 5 & 7\\ 9 & -7 & 2\\ 1 & -1 & 0\end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

MMT277

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	A	A,B,C	B,D	2	225	625B	9A

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-05

Single Correct Question

1. If 'A' is a square matrix such that $A^2 = I$ then A^{-1} is equal to -

(A) $A + I$ (B) A (C) 0 (D) $2A$

MMT278

2. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of matrix A, then α is-

(A) -2 (B) -1 (C) 2 (D) 5

MMT279

3. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det.(A^T A^{-1})$ then which of the following can not be the value of $\underbrace{f\left(f\left(f\left(f.....f(x)\right)\right)\right)}_{n \text{ times}}$ is $(n \geq 2)$

(A) $f^n(x)$ (B) 1 (C) $f^{n-1}(x)$ (D) $nf(x)$

MMT280

Rankers Academy JEE

4. If A is a square matrix such that $A^5 = O$ and $A^n \neq O$ for $n = 1,2,3,4$, then $(I - A)^{-1}$ is equal to (where O is a null matrix)-

(A) $I + A$ (B) $I + A + A^2$ (C) $I + A^{-1}$ (D) $I + A + A^2 + A^3 + A^4$

MMT281

5. If A and B are non-singular symmetric matrices of the same order such that $AB = BA$ and $P = (AB^{-1})^T$ & $Q = (A^{-1}B)^T$ then $(PQ)^{-1}$ is equal to -

(A) $AB^{-1}AB$ (B) $A^{-1}B^{-1}AB$ (C) $A^{-1}BA^{-1}B$ (D) A^2B^2

MMT282

Multiple Correct Question

6. Let $A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, then -

(A) $7|A| = \frac{1}{2}$ (B) $|adj\ A| = \frac{1}{196}$

(C) trace $(adjA) = -\frac{1}{7}$ (D) Matrix A is a symmetric matrix

MMT283

For More Join: @IITJEE_Advanced

7. If $A=\begin{bmatrix}3 & 0 & 0\\ 2 & 2 & 0\\ 4 & 5 & 3\end{bmatrix}$, then which of the following is(are) true ?

(trace of A denotes sum of principal diagonal elements of A)

(A) A is invertible (B) trace$(adj(adj(A)))$ = 144

(C) trace $(adj(adj(A)))$ = 8 (D) $|adj\ A|$ is less than 400

MMT284

Integer Type Question

8. If A is a non-zero idempotent matrix satisfying $\left(I-\frac{2}{5}A\right)^{-1}=I-\alpha A$ where I is the unit matrix of the same order as that of A then the absolute value of 3α is

MMT285

9. Let A, B and $A + B$ are non-singular matrices of order 3 × 3 satisfying $A^{-1} + B^{-1} = (A + B)^{-1}$ and $|AB^{-1}| \in R$ then value of $\frac{|A|}{|B|}$ is

MMT286

10. Let P, Q, R be invertible matrices of second order such that $A = PQ^{-1}$, $B = QR^{-1}$, $C = RP^{-1}$, then $\det(ABC + BCA + CAB)$ is

MMT287

11. An invertible matrix A of order 3 satisfies the relation $A = A^{-1} + 2I$, (where I denotes identity matrix). The value of $|A - I|.|A + I|.|A - 2I|$ is

MMT288

Rankers Academy JEE

Subjective Type Question

12. If $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$; $B=\begin{bmatrix}3 & 1\\ 1 & 0\end{bmatrix}$; $C=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$ and $X=\begin{bmatrix}x_1 & x_2\\ x_3 & x_4\end{bmatrix}$, then solve the following matrix equation.

(a) $AX = B - I$ (b) $(B - I)X = IC$ (c) $CX = A$

MMT289

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	D	D	B	B,C,D	A,B,D	2	1	9
Question	11									
Answer	8									

12. (a) $X=\begin{bmatrix}-3 & -3\\ \frac{5}{2} & 2\end{bmatrix}$, (b) $X=\begin{bmatrix}1 & 2\\ -1 & -2\end{bmatrix}$, (c) no solution

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Matrices : RACE-06

Single Correct Question

1. If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$ is zero divisor of $f(x) = x^2 - 4x - 5$, then A^3 is equal to

(A) $\begin{bmatrix}41 & 42 & 42\\ 42 & 41 & 42\\ 42 & 42 & 41\end{bmatrix}$ (B) $\begin{bmatrix}42 & 41 & 42\\ 41 & 42 & 41\\ 44 & 41 & 42\end{bmatrix}$ (C) $\begin{bmatrix}43 & 42 & 41\\ 44 & 41 & 42\\ 42 & 43 & 43\end{bmatrix}$ (D) $\begin{bmatrix}41 & 42 & 41\\ 42 & 41 & 42\\ 42 & 42 & 41\end{bmatrix}$

MMT290

2. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ & all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$, then the value of $(x + y + z)$ where I is a unit matrix of order 3, is -

(A) 0 (B) 1 (C) 2 (D) 3

MMT291

3. The system of equations $x + 2y - 3z = 1$, $x - y + 4z = 0$, $2x + y + z = 1$ has -

(A) only two solutions (B) only one solution

(C) no solution (D) infinitely many solutions

Rankers Academy JEE

MMT292

4. The system of equations $x + y + z = 8$, $x - y + 2z = 6$, $3x + 5y - 7z = 14$ has-

(A) Unique solution (B) infinite number of solutions

(C) no solution (D) none of these

MMT293

5. Consider the system of equations $x - 2y + 3z = -1$, $x - 3y + 4z = 1$ and $x + y + 2z = k$ has exactly three solutions, then -

(A) $k = 3$ (B) $k = 0$ (C) $k = -9$ (D) $k = \phi$

MMT294

Integer Type Question

6. If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$, $x + 4cy + cz = 0$, has a non- trival solution & $a, b, c > 0$ then the minimum value of $\left(\frac{a^3 + c^3}{b^3}\right)$ is

MMT295

7. Square matrices B of second order are formed using some or all the possible values of k, when system of linear equations $kx + 4y + z = 0$, $4x + ky + 2z = 0$ & $2x + 2y + z = 0$, has infinite solution, then number of singular matrices B is

MMT296

For More Join: @IITJEE_Advanced

Subjective Type Question

8. Determine the characteristic roots of the matrix A. Hence find the trace and determinant value of A.

Where $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and also prove that $A^3 - 18A^2 + 45A = \mathbf{O}$.

MMT297

9. Find the characteristic polynomial equation of $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}$

MMT298

10. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

MMT299

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	A	B	D	A	D	2	6
Question	8				9		
Answer	$\lambda = 0,3$ and 15 $tr(A) = 18, \lvert A\rvert = 0$				$x^3 - 6x^2 + 13x - 11 = 0$		

For More Join: @IITJEE_Advanced